1(a)(i)	$(2, 12)$	B1
(ii)	$(2, 15)$	B1
		(2)
(b)	k ,, 3, $k = 5$	M1 A1
		(2)
(c)	$(x =) -5$	B1
		(1)
		(5 marks)

2(a)	$6=5a+b,\ 3.3=8a+b$	M1 A1
	$a=-0.9, b=10.5$	dM1 A1
		(4)
(b)(i)	$10.5\ \text{m}^3$	B1ft
(ii)	$-0.9\sqrt{t}+10.5=0 \Rightarrow 0.9\sqrt{t}=10.5 \Rightarrow t=...$	M1
	$t=136\ (\text{min})$	A1
		(3)
		(7 marks)

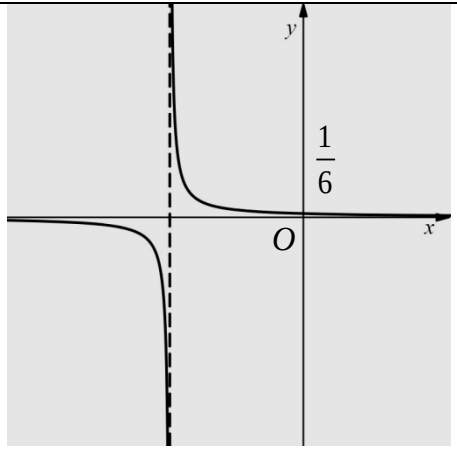
3(a)	$\left(\mathrm{f}(x)=\right)-3\cos x$ or $\left(\mathrm{f}(x)=\right)3\sin(x-90°)$	M1 A1
		(2)
(b)(i)	8	B1
(ii)	5	B1
		(2)
		(4 marks)

4(i)	Uses a correct law of indices on 2^{4k-3} or 8^{1-k} The possibilities are endless but some more common examples are: For 2^{4k-3}: $2^{4k}\times 2^{-3}$, $\dfrac{2^{4k}}{8}$, $8^{\frac{4k-3}{3}}$, $4^{\frac{4k-3}{2}}$, $\left(\sqrt{2}\right)^{8k-6}$ For 8^{1-k}: 8×8^{-k}, $\dfrac{8}{8^{k}}$, $2^{3(1-k)}$, $4^{\frac{2(1-k)}{3}}$, $\left(\sqrt{2}\right)^{6-6k}$ These may be seen in isolation e.g. not in an equation. But not just e.g. $8=2^3$	M1
	e.g. $2^{4k-3}=\dfrac{8^{1-k}}{4\sqrt{2}}\Rightarrow 2^{4k-3}=\dfrac{2^{3(1-k)}}{2^{\frac{5}{2}}}\Rightarrow 2^{4k-3}=2^{3(1-k)-\frac{5}{2}}$ $\Rightarrow 4k-3=3(1-k)-\dfrac{5}{2}\Rightarrow k=...$	dM1
	$k=\dfrac{1}{2}$	A1
		(3)

(ii)	$\dfrac{x\sqrt{3}+2}{\sqrt{3}-1}=x\sqrt{3}-4\Rightarrow x\sqrt{3}+2=\left(x\sqrt{3}-4\right)\left(\sqrt{3}-1\right)$ $\Rightarrow x\sqrt{3}+2=3x+4-4\sqrt{3}-x\sqrt{3}$ $\Rightarrow 2\sqrt{3}x-3x=2-4\sqrt{3}$	M1
	$\Rightarrow x\left(2\sqrt{3}-3\right)=2-4\sqrt{3}$	A1
	$\Rightarrow x=\dfrac{2-4\sqrt{3}}{2\sqrt{3}-3}\times\dfrac{2\sqrt{3}+3}{2\sqrt{3}+3}$	dM1
	$=-6-\dfrac{8}{3}\sqrt{3}$	A1
		(4)
	Alternative	
	$\dfrac{x\sqrt{3}+2}{\sqrt{3}-1}=x\sqrt{3}-4\Rightarrow\dfrac{x\sqrt{3}+2}{\sqrt{3}-1}\times\dfrac{\sqrt{3}+1}{\sqrt{3}+1}=x\sqrt{3}-4$ $\Rightarrow\dfrac{3x+2\sqrt{3}+x\sqrt{3}+2}{3-1}=x\sqrt{3}-4$ $\Rightarrow 3x+2\sqrt{3}+x\sqrt{3}+2=2x\sqrt{3}-8\Rightarrow\sqrt{3}x-3x=10+2\sqrt{3}$	M1
	$\Rightarrow x\left(\sqrt{3}-3\right)=10+2\sqrt{3}$	A1
	$x=\dfrac{10+2\sqrt{3}}{\sqrt{3}-3}\times\dfrac{\sqrt{3}+3}{\sqrt{3}+3}$	dM1
	$=\dfrac{10\sqrt{3}+30+6+6\sqrt{3}}{3-9}=-6-\dfrac{8}{3}\sqrt{3}$	A1

	$y=\frac{1}{2}x^4-3+\frac{10}{x^2}$	
5(a)	$\left(\int \frac{1}{2}x^4-3+\frac{10}{x^2}\,\mathrm{d}x=\right)\frac{1}{10}x^5-3x-\frac{10}{x}+c$	M1A1A1
		(3)
(b)(i)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)2x^3-\frac{20}{x^3}$	M1A1A1
		(3)
(b)(ii)	$2x^3-\frac{20}{x^3}=3\Rightarrow 2x^6-20=3x^3$	M1
	$2x^6-3x^3-20=0\Rightarrow\left(2x^3+5\right)\left(x^3-4\right)=0\Rightarrow x^3=...\Rightarrow x=...$ or e.g. $x^3=a\Rightarrow(2a+5)(a-4)=0\Rightarrow a=...\Rightarrow x^3=...\Rightarrow x=...$	dM1
	$\Rightarrow x=-\sqrt[3]{\frac{5}{2}},\quad \sqrt[3]{4}$	A1A1
		(4)
		(10 marks)

6.	$(x+5)(3x+2)(2x-5)=(3x^2+17x+10)(2x-5)=\ldots$	M1
	$=6x^3-15x^2+34x^2-85x+20x-50$ $\left(=6x^3+19x^2-65x-50\right)$	A1
	$6x^3+19x^2-65x-50=3x^2-33x-50$ $\Rightarrow 6x^3+16x^2-32x=0$	dM1
	$2x\left(3x-4\right)\left(x+4\right)=0$	ddM1
	$x=-4,\ 0,\ \frac{4}{3}$	A1
		(5)
		(5 marks)

7(a)	 $x = -6$	Correct shape and position	B1
		Correct **equation** of vertical asymptote. E.g. $x = -6$ oe e.g. $x + 6 = 0$	B1
		Correct y intercept $\left(0, \frac{1}{6}\right)$	B1
			(3)

(b)(i)	$mx-4=\dfrac{1}{x+6}\Rightarrow\left(mx-4\right)\left(x+6\right)=1$	M1
	$\Rightarrow mx^2+(6m-4)x-25=0$ or $mx^2+6mx-4x-25=0$	A1
	$"b^2-4ac"=(6m-4)^2-4\times m\times\left(-25\right)$	dM1
	$\Rightarrow(6m-4)^2+100m\ldots0$ $\Rightarrow 36m^2+52m+16\ldots0\Rightarrow 9m^2+13m+4\ldots0$ *	A1*
(ii)	$\left(m+1\right)\left(9m+4\right)\ldots0\Rightarrow\quad m,,-1,\quad m\ldots-\dfrac{4}{9}$	M1 A1
		(6)

8(a)	$m = \dfrac{18-0}{2--3}$	M1
	$y-18=\dfrac{18}{5}(x-2)$ or e.g. $y=\dfrac{18}{5}(x+3)$ or $18=\dfrac{18}{5}(2)+c \Rightarrow c=...$ or e.g. $0=\dfrac{18}{5}(-3)+c \Rightarrow c=...$	dM1
	$y=\dfrac{18}{5}x+\dfrac{54}{5}$	A1
		(3)
(b)	Examples: $y=ax^2+bx-6$ $y=(x+3)(px+q)$ $y=ax^2+bx+c,\ (0,-6),\ (2,18),(-3,0) \Rightarrow a=...$ or $b=...$ or $c=...$ NB $a=\dfrac{14}{5},\ b=\dfrac{32}{5},\ c=-6$	M1
	Full method to find all constants. E.g. Solves $9a-3b-6=0$ and $4a+2b-6=18 \Rightarrow a=..., b=...$	dM1
	Either $a=\dfrac{14}{5}$ or $b=\dfrac{32}{5}$ $y=\dfrac{14}{5}x^2+\dfrac{32}{5}x-6$	A1 A1
		(4)
(c)	$\dfrac{14}{5}x^2+\dfrac{32}{5}x-6<y<\dfrac{18}{5}x+\dfrac{54}{5}, \quad x<0$	M1 A1ft
		(2)
		(9 marks)

9(a)	$\left(CQ^2 =\right)0.5^2+1.84^2-2\times0.5\times1.84\cos0.8$	M1
	(Radius =) $CQ=1.534\,\text{m}$	A1
		(2)
(b)	$\dfrac{\sin PCQ}{0.5}=\dfrac{\sin 0.8}{"1.534"}$ or $0.5^2="1.534"^2+1.84^2-2\times"1.534"\times1.84\cos PCQ$	M1
	$\Rightarrow \sin PCQ=\dfrac{0.5\sin0.8}{"1.534"}\left(=0.233....\right)\Rightarrow PCQ=0.236*$ or $\cos PCQ=\dfrac{"1.534"^2+1.84^2-0.5^2}{2\times"1.534"\times1.84}\left(=0.972...\right)\Rightarrow PCQ=0.236*$	A1*
		(2)
(c)	$\dfrac{1}{2}r^2\theta=\dfrac{1}{2}\times"1.534"^2\times\left(2\pi-0.236\right)$ oe e.g. $\pi"1.534"^2-\dfrac{1}{2}\times"1.534"^2\times0.236$ (= 7.114…) OR $\dfrac{1}{2}ab\sin C=\dfrac{1}{2}\times0.5\times1.84\sin\left(0.8\right)$ or $\dfrac{1}{2}\times"1.534"\times1.84\sin\left(0.236\right)\left(=0.3299...\right)$	M1
	Attempts $\dfrac{1}{2}r^2\theta=\dfrac{1}{2}\times"1.534"^2\times\left(2\pi-0.236\right)$ **AND** $\dfrac{1}{2}ab\sin C=\dfrac{1}{2}\times0.5\times1.84\sin\left(0.8\right)$ **AND** adds	dM1
	(awrt) 7.4 (m^2)	A1
		(3)
(d)	Attempts $r\theta="1.534"\times\left(2\pi-0.236\right)\left(=9.276...\right)$	M1
	Perimeter = $"1.534"\times\left(2\pi-0.236\right)+0.5+\left(1.84-1.534\right)=10.1$ (m)	A1
		(2)
		(9 marks)

10(a)	$m = \frac{4}{5}$	B1
	$y+3=\frac{4}{5}(x-4)$ or $-3=\frac{4}{5}\times 4+c \Rightarrow c=...$	M1
	$y=\frac{4}{5}x-\frac{31}{5}$	A1
		(3)
(b)	$f'(4)=\frac{4}{5} \Rightarrow \frac{k\sqrt{4}(4-3)}{5}=\frac{4}{5} \Rightarrow k=...$	M1
	$k=2$	A1
		(2)
(c)	$(f'(x)=)\frac{2\sqrt{x}(x-3)}{5}=\frac{2x^{\frac{3}{2}}}{5}-\frac{6x^{\frac{1}{2}}}{5}$	M1
	$(f(x)=)\frac{4x^{\frac{5}{2}}}{25}-\frac{4x^{\frac{3}{2}}}{5}(+c)$	M1 A1ft
	$x=4, y=-3 \Rightarrow -3=\frac{4(4)^{\frac{5}{2}}}{25}-\frac{4(4)^{\frac{3}{2}}}{5}+c \Rightarrow c=\left(-\frac{43}{25}\right)$	ddM1
	$(f(x)=)\frac{4x^{\frac{5}{2}}}{25}-\frac{4x^{\frac{3}{2}}}{5}-\frac{43}{25}$	A1
		(5)
		(10 marks)